

Biconomy

AI & Blockchain Fintech Innovation

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Problem and Opportunity

WHY AI CAN DECIDE IN MILLISECONDS BUT EXECUTION STILL TAKES MINUTES, AND HOW BICONOMY CLOSES THE GAP

01

Fragmented Execution

A single cross-chain DeFi operation can require five to eight manual approvals across multiple wallets

02

Unpredictable Gas

Users must hold native tokens on every chain they transact on, making cost-stable automation almost impossible.

03

EOA Wallet Limits

Externally Owned Accounts lack native support for delegated permissions, spending limits, or scoped authorizations.

Neither Alone Solves It

AI ALONE
Identifies the optimal strategy but cannot execute without human approval or unsafe custody.

BLOCKCHAIN ALONE
Provides settlement but cannot adapt, interpret intent, or assess risk dynamically.

These failures compound into one architectural problem: more transactions → more gas exposure → delegating to an AI agent becomes commercially untenable.

WHO IS AFFECTED?

Banks: Cannot integrate intelligent services with legacy rails	Asset Managers: Forced choice between manual overhead and unsafe custody	SMEs: Excluded from sophisticated treasury and cross-border tools	Compliance: Cannot audit off-chain AI decisions producing on-chain outcomes	End Users: Fragmented experience of wallets, gas tokens, and approvals
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Why Now?

\$100B+ DeFi TVL, mostly still managed manually	\$60B+ Projected AI-fintech market by 2031	20M+ ERC-4337 ops (Biconomy helped pioneer)	MiCA + SEC Demanding auditable, transparent execution
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Sources: Ifeanyichukwu and Vaswani (2025); Gong (2026); Xue et al. (2023); Tseng, Chen, and Yuan (2026); Stoica (2023); Market.us (2024); ESMA (2023); Lin et al. (2024).

Biconomy: The Foundational Layer for Agentic Commerce

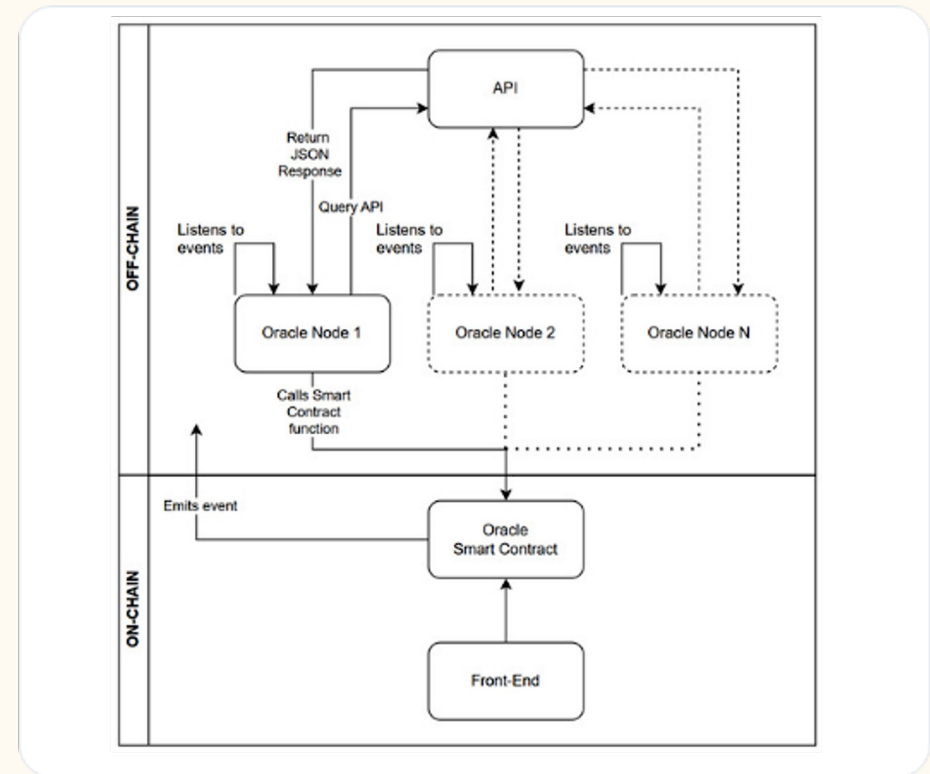
BRIDGING OFF-CHAIN INTELLIGENCE TO ON-CHAIN EXECUTION

Separation of Concerns: AI "thinks" off-chain (computation/reasoning) while **Biconomy** "executes" on-chain (settlement/custody).

ERC-4337 Smart Accounts: Transforms basic wallets into programmable accounts with built-in spending limits and risk guardrails.

Session Keys: Enables AI agents to perform pre-approved transaction batches without requiring continuous human signatures.

Gas Abstraction: Paymasters remove friction by allowing agents to operate gaslessly, critical for autonomous micro-transactions.

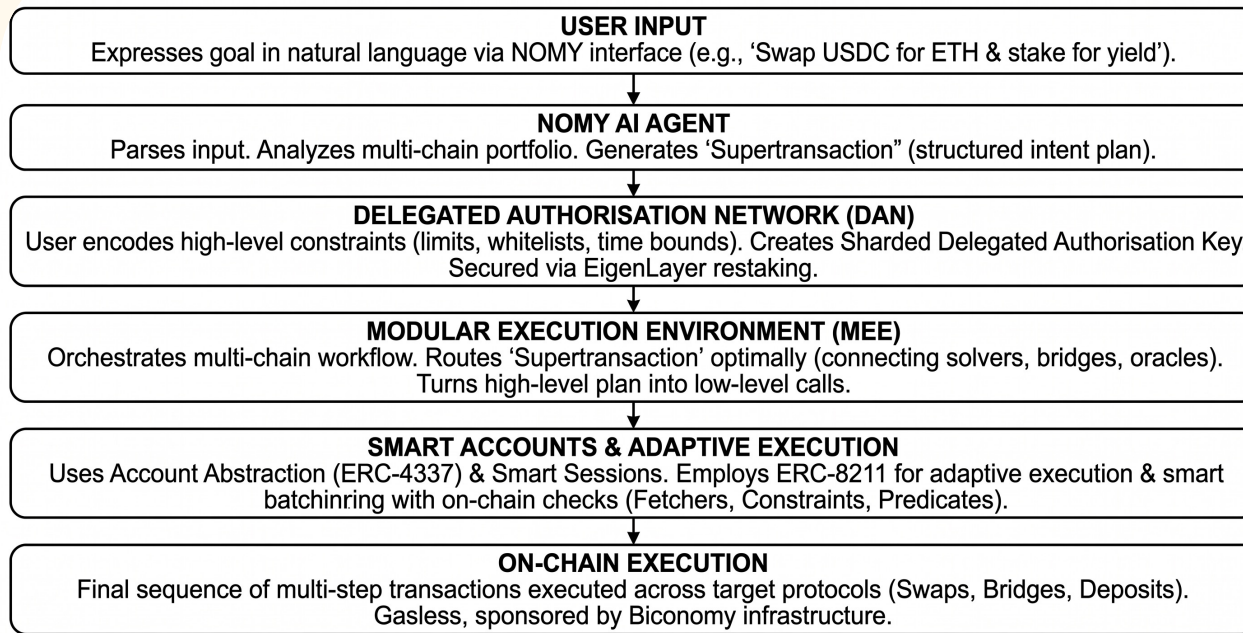


BICONOMY INFRASTRUCTURE

AI Layer

User Intent AI Agent Execution

BICONOMY AI WORKFLOW: FROM USER INTENT TO SECURE EXECUTION



NOMY

Intent Interface

Users chat to convey their goals, then the agent decomposes each intent into a sequence of actions, such as swaps, bridges, approvals, and deposits.

Delegated Authorisation Network (DAN)

This layer allows users to grant AI agents specific powers, with permissions enforced by EigenLayer's restaked validators.

Modular Execution Environment (MEE)

The Modular Execution Environment is an orchestration engine that executes NOMY's AI-generated super transactions on the user's behalf.

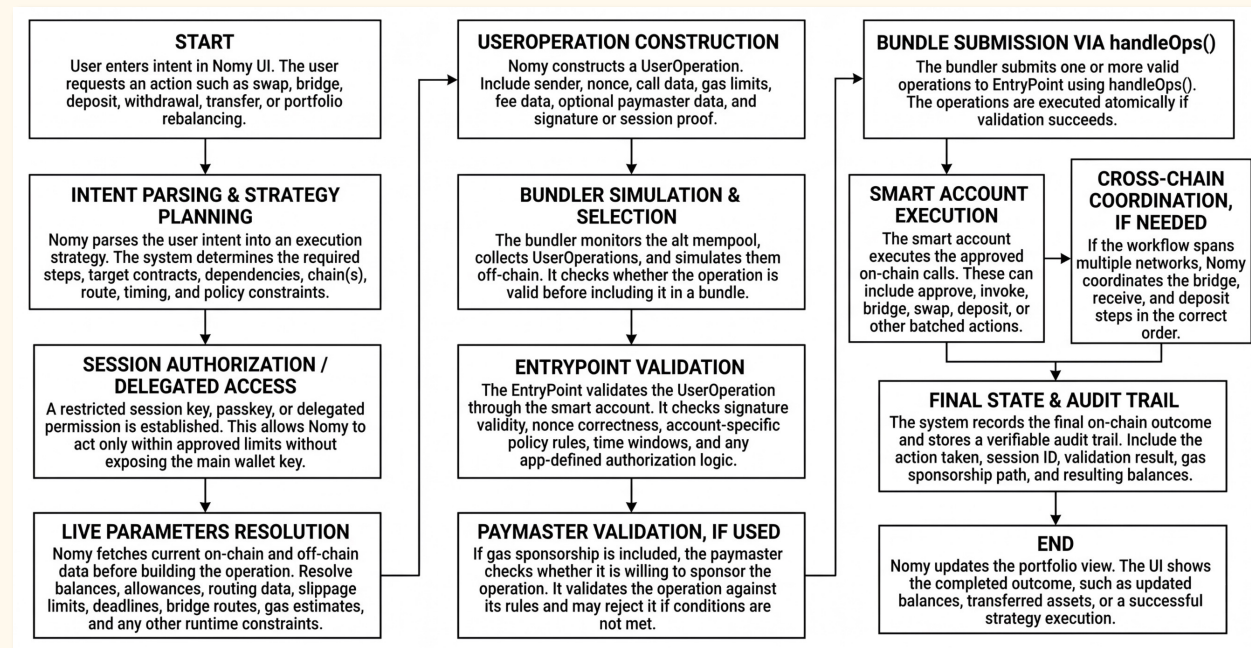
Smart Features, Execution & ERC-8211

Smart sessions allow users to time- and scope-limit AI agents, approving only certain operations during a specified period. Users can set spending limits, whitelists, and revocation rules.

Blockchain Layer

- **Biconomy**'s blockchain uses an ERC-4337-based account abstraction stack comprising Smart Accounts, Bundlers, Paymasters, and Smart sessions.
- Smart Accounts have custom validation rules that are checked before any operation goes through. The validated logic in each account enforces these policy constraints.
- Smart Session, a user can add a session module to their Smart Account. Each session has its own policies that control a variety of parameters. If any rule is broken, validation fails, and nothing changes on the blockchain.
- **Biconomy** uses Paymasters and Bundlers to handle gas. Paymaster contract can cover gas costs or accept ERC-20 tokens as payment Bundlers collect many UserOperations and send them to the shared EntryPoint contract.
- ERC-8211 lets agents describe a series of actions as an off-chain script, which is checked against the current on-chain state when it runs.

Example Future Transaction through **Biconomy**

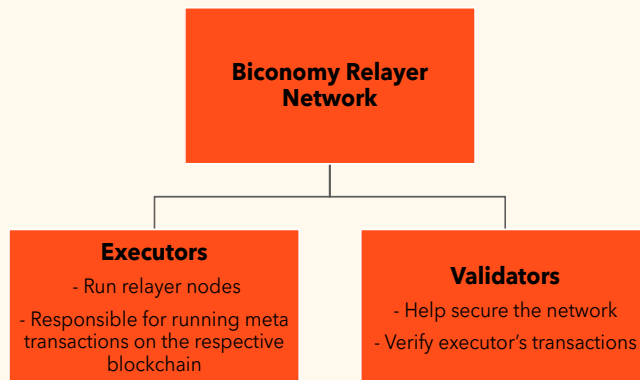


Main Products

Three main products of **Biconomy** are: Mexa, Forward, Hyphen, each one of them solving challenges like onboarding/UX, gas fees, and interoperability.

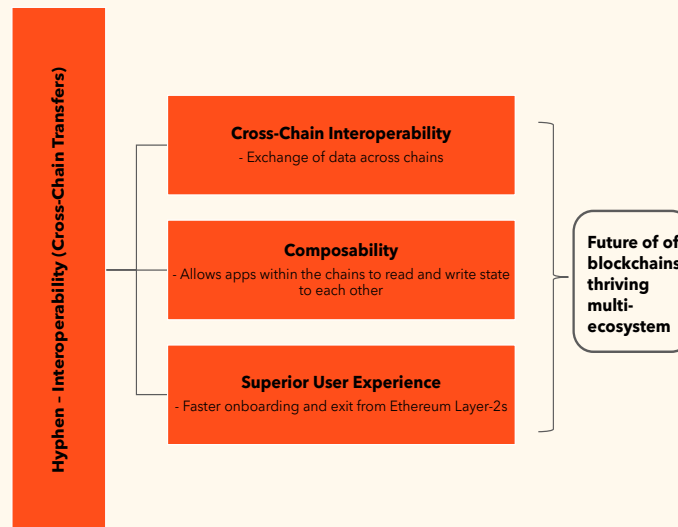
1. Mexa

- JavaScript based, easily integrated into a dAPP
- Supports gasless transactions through **Biconomy** relay network



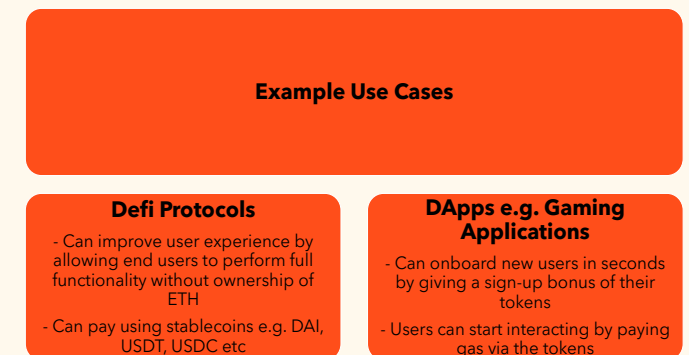
2. Hyphen

- Supports interoperability, cheaper and faster token transfer across blockchain
- Liquidity maintained on both ends of the chain



3. Forward

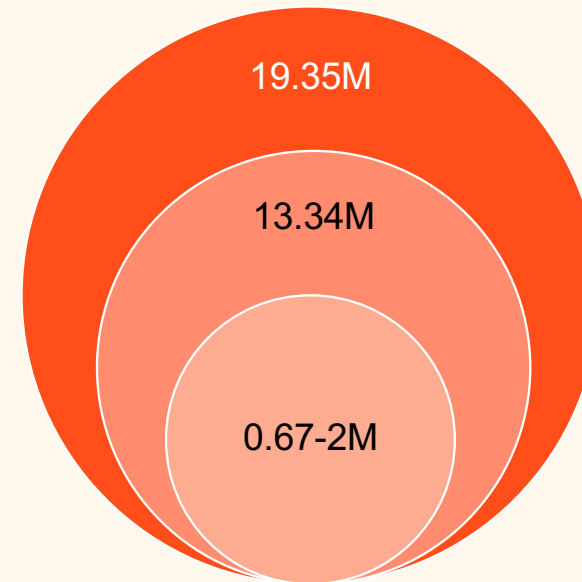
- Allows users to pay transaction fees with any ERC-20 tokens
- Relayers automatically detect congestion, increase gas fees to maintain transaction flow
- Improved user experience.



Market Opportunity & Sizing

- **Biconomy** provides developer infrastructure for cross-chain orchestration, multi-chain execution, and user abstraction, which makes cross-chain transactions the most relevant unit of analysis.
- The interoperability data shows that 11,285,753 cross-chain transactions moved over 28 billion USD across five protocols on 11 blockchains during the last seven months of 2024, equating an annualized transaction TAM of approximately 19.35 million transactions.
- Given that 68.9% of observed cross-chain transactions occur between Layer 2 networks, the serviceable market is concentrated in L2-oriented cross-chain flows.
- A realistic serviceable obtainable market for **Biconomy** can then be derived by applying an adoption-based capture rate to this segment.

Top-Down Market Sizing (USD)



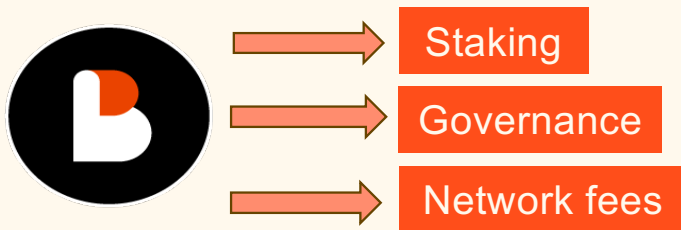
Metric	Definition	Example Estimate
TAM	Annual cross-chain transactions in the addressable market.	19.35m
SAM	Cross-chain transactions matching Biconomy's scope, especially L2-oriented flows.	13.34m
SOM	Portion of SAM Biconomy can realistically capture.	0.67m to 2m

Business Model

How **Biconomy** makes money (B2B2C):

Customers: Web3 developers, dApps, protocols, and financial institutions. Aave, J.P. Morgan Onyx, Mercedes-Benz Acentrik, Gemini, dYdX, PancakeSwap.

4 Revenue Streams: Paymaster fees (gas sponsorship + revenue share with high-volume partners), Bundler-as-a-service (ERC-4337), SDK and API access (volume-based pricing), MEE orchestration fees (paid in BICO).



Unit economics: Revenue $\approx$ Take Rate $\times$ Transactions per Account $\times$ Active Accounts. Marginal cost $\approx 0 \rightarrow$ operating leverage of a payment processor, not a SaaS company.

Traction: 4M+ users, 400+ dApps, 17M+ transactions, \$1.4B+ volume, ~\$10.5M raised (Seed $\rightarrow$ Series B; Bain Capital, Coinbase Ventures, Jump Capital).



AI agents already drive ~28% of transactions on Gnosis. Same fees, just transacted far more often.

Traction & Team

Rapid Growth of **Biconomy** & Team behind it

- The traction includes more than 4 million end users onboarded through the SDK, over **400** integrated dApps, more than **17 million** processed transactions, and over **USD \$1.4 billion** in historical transaction volume.



- Biconomy** has raised \$10.5M across seed, Series A, and Series B stages.
- The founding team has technical depth, but the business remains highly founder-led, with key-person dependency and moderate risk from concentration within the original core team. (Appendix A)

Metric	Value	Context
Unique Users Onboarded	> 5 million	As of 2024 via AA-SDK (Decrypt, Mar 2024)
Smart Accounts Created	> 1.5 million	Via Biconomy infrastructure (Decrypt, Mar 2024)
UserOps Processed	> 6.1 million	ERC-4337 UserOperations (Decrypt, Mar 2024)
dApps Integrated	> 400	Across 20+ EVM chains (Decrypt, Mar 2024)
Chains Supported	20+ EVM chains	Including Scroll, Blast, ZetaChain
AA Market Share (est., 2023)	8–50 %	Paymaster vol: 8%; UOps share 50% (BlockEden)
System Uptime Claim	99.9 %	Enterprise-grade SLA (CoinMarketCap)
Gas Tokens Supported	> 10,000 ERC-20	Users can pay gas in any ERC-20 token

Biconomy Founding Team

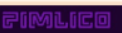


Competitors and Competitive Positioning

The Challengers to **Biconomy**

- **Biconomy's** core differentiators are its transaction abstraction, programmable smart account infrastructure, and user-centric automation designed to support seamless Web3 and AI-agent interactions.
- Its main competitors like ZeroDev or Pimlico, which primarily focus on infrastructure scale, institutional wallet security, or exchange ecosystems. (Appendix B)
- **Biconomy's** differentiation is strongest where usability matters most, particularly in simplifying complex multi-step blockchain interactions for both users and developers.

Comparable Companies

Comparable Companies	What they do well
 Alchemy	Expanding developer ecosystem, dependable infrastructure, enterprise scale
 Safe	Reliable smart account security, multisig dominance
 Coinbase	Large distribution with regulatory credibility
 ZeroDev	Robust developer-first ERC-4337 tooling
 Pimlico	Offers a specialized version of ERC-4337 within its blockchain stack
 Stackup	Developer-friendly account abstraction APIs
 thirdweb	Simplified Web3 onboarding and SDK ecosystem

Risks & Mitigation Techniques

Category	Key Risks	Current Risk Likelihood	Mitigation Techniques	Future Risk Impact
Regulatory	1. Regulatory uncertainty due to overlap between MiCA, GDPR, AI governance and crypto regulations	High	Maintain a "compliance-by-design" architecture, integrate KYC/AML protocols, and ensure AI-agents support execution rather than investment advice	Medium
	2. Indirect regulatory exposure from third-party dApps using Biconomy Infrastructure	Medium	Require dApps to disclose regulatory status and strengthen compliance partnerships across jurisdictions	Low
Privacy and Data Security	1. Exposure to GDPR and automated decision-making requirements	High	Use off-chain encrypted processing, privacy impact assessments and strict access controls	Medium
	2. Re-identification Risk	Medium	Minimise personal Data collection and avoid storing sensitive data on-chain	Low
AI Bias and Model Reliability	1. AI models producing biased or inaccurate decisions	High	Implement AI Governance	Medium
	2. AI models failing during volatile DeFi market conditions	Medium	Introduce human oversight, transaction limits and pre-defined risk controls for AI agents	Low
Adoption and User Education	1. Users underestimating AI-agent permissions and automated transactions	Medium	Create transaction previews and clearer explanations of transaction consequences	Low
Operational Complexity and Scaling	1. Downtime and congestion	High	Build redundancy into bundler and paymaster infrastructure and introduce fallback systems	Medium
Smart Contracts and Technical Execution	1. Coding vulnerabilities exposing users to permanent asset loss	High	Conduct independent smart contract audits and staged deployment testing	Medium

Sources: Biconomy (2026); ESMA (2026); KPMG (2022); GDPR (2017); EUROFI (2024); Biconomy (2026); Biconomy (2026); Biconomy (2026); Biconomy (2026)

Biconomy Implementation Timeline

Seven years of infrastructure, ending in a co-authored Ethereum standard

Implementation Timeline

	2019-2021	June 2024	Nov-Dec 2024	Aug 2025	April 2026
Founding + meta-transactions					
DAN launch					
Klaster acquisition + Nomy					
Gemini Wallet on Nexus					
ERC-8211 with Ethereum Foundation					

Future Direction (Next 12-24 Months)

Standards

ERC 8211 + EIP 8141

If both ship, every Ethereum account gets smart-account capabilities by default.
Biconomy becomes a baseline, not an opt-in.

Distribution

Nexus Ecosystem Expansion

Positioned as the 'OP Stack of smart accounts'. Gemini is the lead institutional partner; more wallets are likely to follow.

Open Question

BICO Token Value Capture

Adoption is climbing, the token is not. The forward question is whether the team can re-link protocol usage to token value before investors stop expecting them to be linked

Sources: YourStory (2019); Yahoo Finance (2024); Biconomy (2024); Biconomy (2025a); Biconomy (2025b); The Defiant (2026); CoinMarketCap (2026); Appendix G / Appendix A report.

What changes if this works

HUMANS WILL SET OBJECTIVES. AGENTS WILL EXECUTE. INFRASTRUCTURE IS THE CONSTRAINT.

Why Incumbents will not build this Layer

Banks

No technical posture, no commercial incentive to build rails that disintermediate them.

Crypto Wallet Providers

Optimized for human users, not autonomous agents, designed to be signed by people.

Biconomy

Sits between these two camps. Already shaping ERC-4337 and the building blocks of autonomous finance.

What Changes if this Works

Today

WITH PROGRAMMABLE EXECUTION

Humans operate
financial tools



Humans set
objectives for AI
agents

Manual approval per
transaction



Autonomous
execution within
scoped permissions

Fragmented multi-
chain workflows



Unified cross-chain
execution

Trust intermediated
by institutions



Trust enforced
programmatically

Conclusion

Biconomy is a credible execution layer for autonomous finance, but questions still remain

- **Biconomy** has created a programmable financial environment that supports delegated permissions, auto-execution, and cross-chain coordination.
- Open questions remain like in the BICO token utility; which raises questions about whether the team can build a clearer link between protocol usage and token value.
- This analysis concludes that **Biconomy** is a credible execution layer for autonomous finance, but long-term success depends on the broad adoption of AI agents and sustain crypto usage.

WHAT BECOMES OPERATIONALLY VIABLE

Machine-to-Machine Payments Autonomous transfers between AI agents	Autonomous Treasury AI-managed liquidity and yield routing
AI-Managed Investing Continuous strategies within guardrails	Real-Time Cross-Chain Single programmable action across chains

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Appendix A: Team Analysis

Team dimension	Evidence from Biconomy	What it signals	Assessment
Founder experience	Biconomy was founded in 2019 by Ahmed Al-Balaghi, Aniket Jindal, and Sachin Tomar. Ahmed is the CEO, Aniket is COO, and Sachin is CTO.	The founding team has stayed intact and has clear functional role separation.	Strong. Stable founding structure is a positive signal for execution and governance.
Technical depth	Sachin Tomar is listed as Co-Founder and CTO. He has an engineering background and product stack spans smart accounts, paymasters, bundlers, DAN, and MEE.	The team appears capable of building deep infrastructure, not just a wrapper product.	Strong. The technology is complex, so technical leadership matters materially.
Domain relevance	The company focuses on Web3 transaction infrastructure, account abstraction, and AI-agent execution. The founding team is consistently presented as blockchain-native and web3-focused.	The team is operating inside a domain where product-market fit depends on niche technical knowledge.	Strong. This is a domain where generalist product teams are weaker.
Complementary roles	The top team separates CEO, COO, CTO, and people leadership, with advisors including Matthew Liu and Sandeep Nailwal.	Indicates a broader operating structure beyond the founders alone.	Positive. The mix suggests early institutionalization and external credibility.
Execution evidence	Biconomy claims more than 4 million end users, 400+ integrated dApps, 17 million+ processed transactions, and over US\$1.4 billion in historical transaction volume.	Strong proof that the team has moved beyond concept into deployment.	Strong. Traction reduces “idea-only” risk.
Advisor quality	Advisors listed include Matthew Liu and Sandeep Nailwal. Matthew Liu is a known crypto founder, and Sandeep Nailwal is a major Web3 figure.	High-quality advisors can improve network access, credibility, and strategic guidance.	Positive. Particularly useful in crypto, where trust and network effects matter.
Team scale and maturity	Public directories show a broad team across product, growth, engineering, partnerships, and people functions.	The company appears to have moved past a three-founder startup into a more mature operating team.	Positive. That supports scalability and go-to-market execution.
Risk / concentration	The business still appears highly founder-led, and the most strategically important roles remain concentrated around the original core team.	Key-person dependency may still be meaningful.	Moderate risk. Strong founders, but execution may still be concentrated.

Appendix B: Porter's 5 Forces

